Autonomous Data Modeling: Multi-Agent Systems for 3NF Normalization and Schema Generation

Modelado de Datos Autónomo: Sistemas Multi-Agente para Normalización 3NF y Generación de Esquemas

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—autonomous agents, multi-agent systems, database normalization, third normal form (3NF), schema generation, data dictionary, large language models (LLMs), PostgreSQL.**

I. Introduction

The design of relational databases consists of translating business requirements, usually expressed in natural language, into a physical schema able to support system operation with guarantees of integrity, consistency, and performance. The foundations of this process were established by Codd with the relational model [1] and consolidated through the formal definition of functional dependencies and normal forms [2], while the entity-relationship model [3] provided the semantic bridge between the real world and the logical structure. Within this framework, achieving the Third Normal Form (3NF) is not a cosmetic goal: it eliminates insertion, update, and deletion anomalies and enables the construction of coherent, traceable data dictionaries.

In practice, however, professional workflows fall short of this ideal. From requirements gathering to DDL code authoring, the designer must identify entities and attributes, determine functional dependencies, detect transitive dependencies, decompose relations, and document every column in a data dictionary. Each of these stages is prone to human error, and commercial CASE tools such as TOAD Data Modeler assist graphical representation and partial code generation but do not automate the semantic decision that guarantees normalization. In complex specifications —with dozens of entities, composite keys, and chains of transitive dependencies— cognitive load and schedule pressure yield schemas that violate 3NF, accumulate redundancy disguised as performance, and produce incoherent data dictionaries whose cost is deferred to production-time migrations and fixes.

The emergence of large language models (LLMs) has opened a promising avenue for automating this life cycle. Recent work has shown that LLMs can derive conceptual models and preliminary schemas from natural-language specifications [4], and that multi-agent architectures can generate high-quality relational schemas from requirements [5]. At the same time, the literature documents clear limits: Kohita [6] showed that the normalization level of a schema systematically affects the accuracy with which LLMs generate SQL; Kanchinadam *et al.* [7] demonstrated that structurally different variants of the same conceptual model produce divergent answers in Text-to-SQL tasks; and ERBench [8] provided evidence that LLMs commit logical errors when inferring relationships among entities, even in simple schemas. These findings converge on a conclusion: reliable schema generation cannot be delegated to a single prompt; it requires explicit and continuous verification of functional dependencies.

In the multi-agent realm, contemporary evidence confirms that specializing roles reduces logical errors. AttributeForge [9] orchestrates 43 specialized agents to model product schemas in production, while GalaxyWeaver [10] shows that an agent ecosystem can iteratively evaluate a deficient schema and autonomously restructure it until an optimal state is reached. These antecedents suggest that the same strategy can be applied to the relational normalization problem, provided that an agent is devoted to auditing the mathematical dependencies that underlie the schema.

This article presents an empirical framework for au-

tonomous data modeling composed of specialized agents: an analyst agent extracts entities, attributes, and relations from the requirements text; a generator agent builds the data dictionary; a coding agent authors the DDL code for PostgreSQL; and an orchestrator agent evaluates the functional dependencies and iterates over the schema until compliance with 3NF is demonstrated before the final DDL is emitted. The contributions of the article are: (i) the definition of a multi-agent architecture with separated cognitive roles for requirements extraction, documentation, and coding, audited by a dependency verifier; (ii) a verification protocol that operates under Codd's formal normalization criteria [2] as the success metric; (iii) an empirical evaluation on complex real-world specifications measuring the system's success rate in producing anomaly-free 3NF schemas; and (iv) a quantitative comparison of design time against the manual process using TOAD Data Modeler.

The remainder of this article is organized as follows. Section II presents the normalization theoretical framework that supports the system. Section III describes the framework architecture and the interaction protocols among agents. Section IV details the experimental protocol and test cases. Section V analyzes the obtained results, and Section VI provides the conclusions.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

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III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
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$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not "Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

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Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

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- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
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- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [17].

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Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.



Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks ...”. Instead, try “R. B. G. thanks...”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

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